

PRISM: Physics-Radio Integrated Simulation and Management for V2X Digital Twins

Chia-Chuan Chiu*, *Student Member, IEEE*, Ming-Chun Lee*, *Member, IEEE*,
Yung-Sheng Chao†, and Li-Chun Wang*, *Fellow, IEEE*

*College of Electrical and Computer Engineering, National Yang Ming Chiao Tung University, Taiwan

†National Chung-Shan Institute of Science and Technology, Taiwan

Email: {steve.ee09, mingchunlee}@nycu.edu.tw, jimys.chao@gmail.com, wang@nycu.edu.tw

Abstract—This paper presents a physics-radio integrated simulation and management (PRISM) platform for vehicle-to-everything (V2X) digital twins. PRISM is built on a modular co-simulation framework that integrates ray tracing for wireless propagation with vehicle motion simulation, enabling generation of spatial-temporal aligned multimodal data for V2X scenarios. In addition, PRISM employs a context-driven approach for information and control exchange, where an orchestrator module coordinates context data exchange and command control among distributed modules. Furthermore, to ensure simulation fidelity, PRISM addresses the sim-to-real gap problem in ray tracing caused by imperfect antenna modeling through gradient-based antenna pattern reconstruction. Experimental results show that PRISM can provide near real-time simulations across varying scenarios, enabling deployment for large-scale V2X simulations in practical systems. Additionally, a closed-loop beam selection use case is developed, in which an external control module within PRISM seamlessly integrates simulation results to enable context-aware beam switching. This use case effectively validates the usefulness and extensibility of PRISM in enhancing system performance.

Index Terms—Wireless digital twin, vehicle-to-everything (V2X), differentiable ray tracing, antenna pattern reconstruction, multimodal simulation

I. Introduction

The evolution toward 6G wireless systems brings considerable challenges in network optimization, particularly for vehicular communication scenarios where millimeter-wave propagation, high mobility, and dense urban environments lead to highly dynamic channel conditions. Integrated sensing and communication (ISAC), a key enabler for 6G applications such as autonomous driving and vehicle-to-everything (V2X) networks [8], requires more accurate modeling and real-time predictions for both wireless propagation and physical environments.

Digital twins are recognized as a critical solution to address the challenges [9]. By creating high-fidelity virtual replicas of wireless environments, digital twins enable rapid generation of diverse data for AI model training while eliminating the

risks and costs associated with real-world experiments. Furthermore, digital twins can also be integrated into real-time systems for real-time simulations of environments to enhance the decisions and operations. However, existing digital twin frameworks for V2X networks face three critical limitations. First, tightly-coupled module integration restricts extensibility, requiring substantial code modifications when adding new modules. Second, the lack of standardized control interfaces hinders integration with AI agents for autonomous optimization. Third, imperfect electromagnetic characterizations such as antenna modeling cause significant sim-to-real gaps [10].

To address these issues, we in this work propose a physics-radio integrated simulation and management (PRISM) platform to create a multimodal digital twin that employs a context-driven modular architecture for V2X simulation. PRISM enables precise co-simulation of wireless propagation, physical environment and vehicle mobility by synchronizing Sionna's [11] ray tracing with CARLA's [12] vehicle dynamics through a loosely-coupled, command-based architecture. Inspired by the model context protocol (MCP) [13], PRISM's orchestrator coordinates different modules via structured JavaScript object notation (JSON) commands, enabling distributed deployment, runtime reconfiguration, and extensibility for autonomous optimization with a context-driven manner. Indeed, this architectural design transforms digital twins from static simulation tools into programmable platforms for dynamic network management.

Beyond architectural innovation, ensuring simulation fidelity remains critical for digital twin deployment. Specifically, to address the sim-to-real gap in wireless propagation simulation, PRISM incorporates a gradient-based antenna pattern reconstruction that leverages Sionna's differentiable ray tracing mechanism. This approach requires only limited measurement data to reconstruct antenna patterns, ensuring the simulated results of PRISM to more accurately reflect real-world propagation conditions. In summary, the main technical contributions of this work are as following:

- **Multimodal co-simulation architecture:** A modular framework that synchronizes CARLA's vehicle mobility with Sionna's ray tracing, enabling efficient generation of spatial-temporal aligned multimodal data, such as RGB camera, depth information, mobility, and wireless channel.

This work has been partially funded by the National Science and Technology Council under the Grants 114-2221-E-A49-185-MY3, and 113-2218-E-A49-027-, and 114-2224-E-A49-002-, and 114-2218-E-A49-019-Taiwan. This work was supported by the Higher Education Sprout Project of the National Yang Ming Chiao Tung University and Ministry of Education (MOE), Taiwan.

Table I: Comparison of Digital Twin Frameworks for Wireless Communication

Reference	Ray Tracing	Calibration Approach	Mobility	Multimodal	Application
Ali [1]	✓	Image-based	Unspecified	✗	Channel modeling
Zubow [2]	✓	✗	Stochastic	✗	Indoor/Outdoor WiFi
Borges [3]	✓	✗	AirSim	✓	UAV MIMO beam selection
Gong [4]	✗	✗	GAN-based	✗	Large-scale urban networks
Pegurri [5]	✓	✗	Unspecified	✗	V2V communication
Tang [6]	✗	✗	AirSim	✓	UAV applications
Mao [7]	✓	✗	CARLA	✓	V2X communication
PRISM (Ours)	✓	Antenna pattern	CARLA	✓	V2X communication

- **Context-driven execution approach:** A control approach with MCP-inspired design that enables structured command-based execution that provides human readability, explainability, and large language model compatibility. This design also supports distributed deployment and seamless integration with external modules for autonomous scenario generation and policy optimization in V2X networks.

- **Gradient-based antenna pattern reconstruction:** A machine learning-based method that reconstructs antenna patterns with limited data, ensures simulation fidelity without requiring detailed antenna specifications.

II. Related Work

A. Digital Twin for Wireless Communication

Digital twins are expected to play a crucial role in future wireless communication systems such as mobile networks, smart-city infrastructures, and internet of things (IoT) applications [14], [15]. For V2X and ISAC applications, the digital twin must address two fundamental challenges: (1) Accurate co-simulation of physical and wireless propagation environments. (2) Reducing the sim-to-real gap to ensure simulation fidelity.

Recent researchers have integrated GPU-accelerated ray tracing with network simulators to improve channel modeling in digital twins. Zubow [2] combined Sionna with ns-3 simulator [16] for WiFi scenarios, introducing caching mechanisms to reduce computational overhead. Pegurri [5] created a full-stack V2V communication environment and compares the simulation error between pure ns-3 method and Sionna-assisted method. Notably, the authors adopt a distributed architectural design in which message exchange is handled via UDP (user datagram protocol), providing greater flexibility in deployment.

To achieve higher physical fidelity, several frameworks incorporated vehicle mobility simulators. Borges [3] and Tang [6] utilized AirSim [17] for UAV trajectory simulation. Mao [7] addressed V2X scenarios by combining CARLA with Sionna, generating multimodal datasets for V2X communication research. However, these frameworks do not provide standardized control interfaces for external programs, limiting their ability to integrate with AI agents or autonomous scenario generation.

B. Ray Tracing for Wireless Simulation

Accurate channel modeling is critical for 5G/6G network planning and optimization. Ray tracing techniques, which simulate electromagnetic wave propagation through geometric optics, have received growing attention in recent research due to their ability to capture site-specific multipath characteristics in complex urban environments. However, conventional ray tracing tools such as MATLAB [18] and Wireless InSite [19] suffer from high computational costs, limiting the development for large-scale data generation required by AI-driven wireless research.

Sionna [11] addresses this limitation through GPU-accelerated differentiable ray tracing architecture, achieving computation speeds tens of times faster than the conventional tools [20], [21]. However, these simulation-based approaches face the inherent challenge of sim-to-real gap. Real-world terrain, material properties, and antenna radiation patterns are difficult to replicate perfectly in digital twin environments. Luo [10] demonstrated that even with perfect 3D map modeling, errors in antenna pattern specifications can result in signal strength prediction errors of tens of dBs.

To address the sim-to-real gap, existing calibration approaches have explored optimization of material electromagnetic properties [22]–[25] and image-based radio maps [1]. Despite the improvements in simulation accuracy, these methods remain environment-specific and do not address antenna pattern uncertainties.

C. Summary

Despite significant achievements in wireless digital twin research, several critical issues remain unresolved.

(1) Most existing digital twin architectures for wireless communications lack proper modular design, which is essential for future scalability and AI-driven system control. (2) Current multimodal digital twins assume perfect knowledge of antenna specifications, but antenna patterns are difficult to obtain in practice, leading to inevitable sim-to-real gaps. To address these issues, PRISM introduces a gradient-based antenna pattern reconstruction method to reduce sim-to-real gaps, and employs a context-driven modular architecture to enable co-simulation of wireless and physical environments while supporting future extensibility.

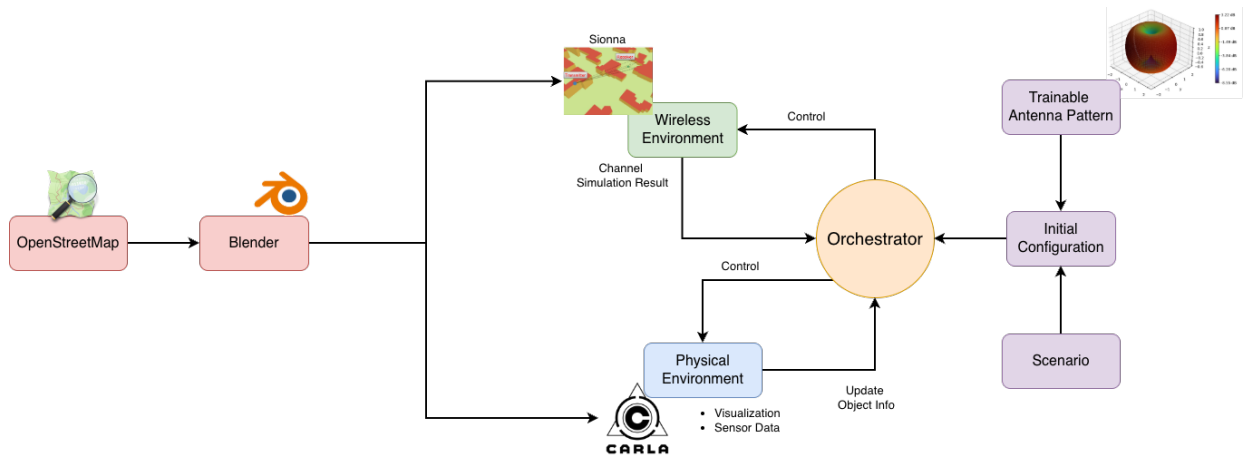


Figure 1: Architecture of PRISM.

III. Proposed Method

A. Overview

Most digital twin frameworks for V2X networks lack standardized control interfaces for runtime reconfiguration and external program integration. This limitation becomes problematic when researchers need to extend the system, such as connecting AI agents to autonomously optimize network performance, or dynamically updating simulation scenarios without a well-defined control protocol. PRISM addresses these problems through a modular co-simulation architecture with a standardized context-driven control interface inspired by MCP. The key design principles of this system are as follows:

- **Modular design with context-driven control for extensibility and usability:** The wireless environment, physical environment, and orchestrator operate as separate modules. Inter-module communication follows a context-driven approach, where structured commands are transmitted via UDP to target modules. This design facilitates distributed deployment across different machines, runtime reconfiguration and integration of new modules into the system. Context-driven approach also ensures human readability, explainability and compatibility with large language models (LLM), enabling natural language interfaces for simulation control.
- **Co-simulation of physical dynamics and electromagnetic environments:** In our digital twin simulation workflow, the orchestrator manages all modules behavior including physical environment and wireless environment. By coordinating the context-driven commands, orchestrator can ensure temporal and spatial synchronization between the two domains.

B. System Implementation

In this section, we describe the construction of our digital twin environment. Figure 1 illustrates the implementation workflow of our digital twin environment.

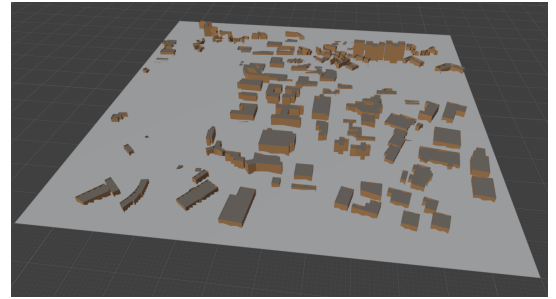


Figure 2: Rendergraph of NYCU campus.

- **Map construction:** To rapidly establish site-specific maps for simulation purposes, we utilized OpenStreetMap [26] to obtain 3D map of specific areas, then converted into formats compatible with Sionna and CARLA separately.
- **Modular design:** To ensure system scalability and usability, we implement a modular architecture for all primary functions. The physical environment module is for simulating vehicle dynamics and outputting the sensor data. The wireless environment module receives object position update commands and executes ray tracing simulation. The orchestrator acts as the central coordinator for sending control commands to all modules. Each module can also operate independently. For instance, the wireless environment module can run on Sionna standalone, while the physical environment module can run on CARLA on its own. Therefore, users may even execute these modules on different devices to run simulations.
- **Context-driven execution:** We adopt a MCP-Inspired context-driven execution method for controlling each module through structured text-based commands. This design enables distributed deployment, runtime reconfiguration, and extensibility for autonomous optimization. Commands are transmitted as JSON-formatted messages via UDP, specifying the target module, operation type,

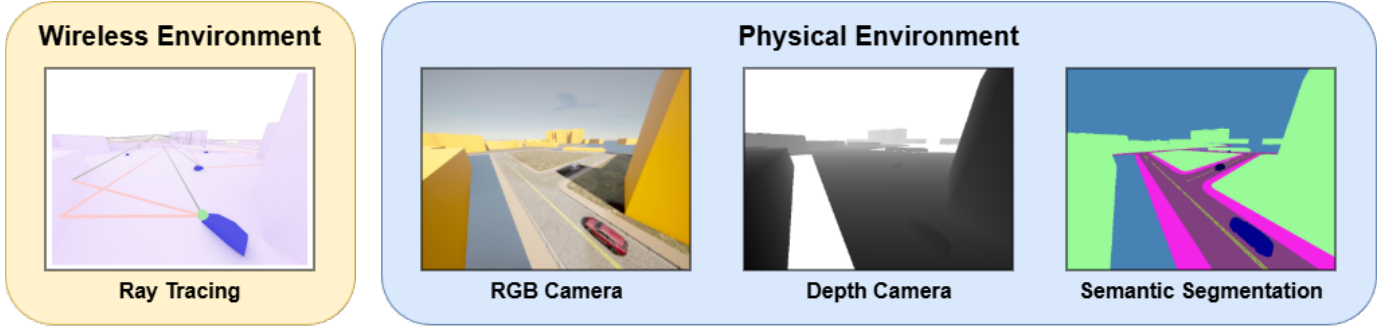


Figure 3: Spatial-temporal aligned multimodal data generation.

and associated parameters. By including object coordinates and simulation timestamps in the commands, spatial-temporal consistency across the physical and wireless environments is naturally maintained. For example, updating a receiver position and triggering ray tracing can be achieved through:

```
{ #command1
  "target": "wireless_env",
  "message_type": "update_object_position",
  "ID": "rx01",
  "position": [10, 20, 1.5]
},
{ #command2
  "target": "wireless_env",
  "message_type": "run_ray_tracing",
  "timestamp": 1
}
```

This approach provides clear semantics for human readability and facilitates programmatic control by external modules.

C. Simulation Workflow

This section describes the complete simulation workflow. First, the orchestrator loads the initialization commands for environment and transmits them to both the wireless environment and physical environment. Subsequently, CARLA simulates urban traffic flow in the physical environment. After each timestamp of simulation, the current positions and orientations of vehicles are recorded and transmitted to the wireless environment via the orchestrator. The status of objects in Sionna are then updated for ray tracing simulation. Figure 1 illustrates how it works in this system. After ray tracing simulation, the orchestrator transmits the information back to the physical environment to begin simulating the next timestamp.

D. Machine Learning-Based Antenna Pattern Reconstruction

The sim-to-real challenge has long been a critical issue in signal simulation. As demonstrated in [10], even with perfect 3D map modeling, errors in antenna hardware modeling, such as incorrect beamwidth settings, can result in signal strength errors of tens of dBs. To address the sim-to-real problem in ray tracing simulation, we developed a machine learning-based

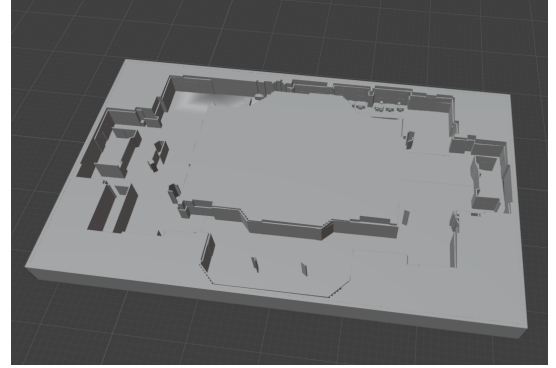


Figure 4: Rendergraph of indoor map.

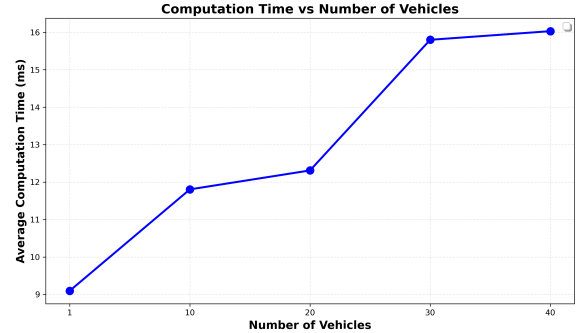


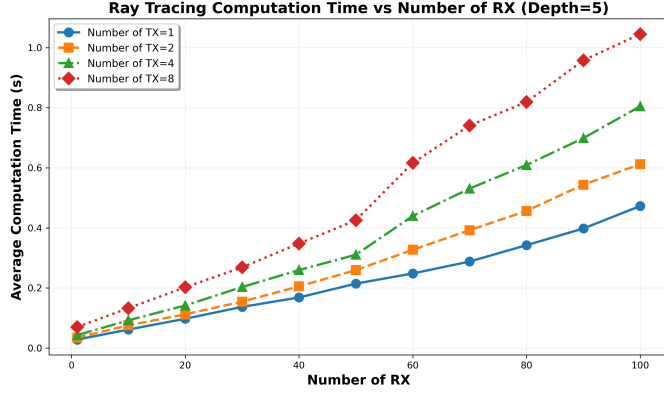
Figure 5: Computation time vs number of vehicles.

antenna pattern reconstruction method leveraging Sionna's differentiable ray tracing capabilities. This approach requires only a limited amount of measured signal strength data to accurately reconstruct antenna patterns, ensuring that the simulated communication data aligns with real-world conditions.

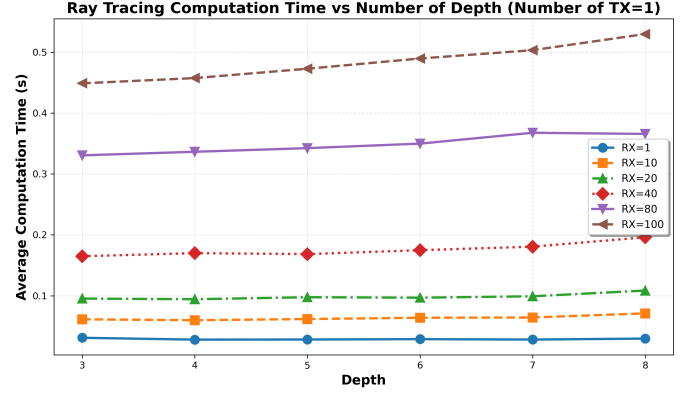
Our reconstruction methodology consists of four steps: (1) Collecting real-world communication data, including positions of transmitter, receiver and channel gains as the dataset. (2) Defining the initial antenna pattern with trainable parameters. (3) Performing ray tracing simulation in the digital twin environment and recording the channel gains. (4) Back propagating the channel gain loss to update the parameters.

Table II: Training progress of antenna pattern reconstruction

Target Pattern	Initial Pattern	Epoch 10	Epoch 50	Epoch 100
3D visualization of the radiation pattern GDB (g)	3D visualization of the radiation pattern GDB (g)	3D visualization of the radiation pattern GDB (g)	3D visualization of the radiation pattern GDB (g)	3D visualization of the radiation pattern GDB (g)



(a) Ray tracing computation time vs number of RX.



(b) Ray tracing computation time vs number of depth.

Figure 6: Ray tracing computation time analysis under different configurations.

IV. Experimental Results and System Validation

In this study, the hardware configuration consists of an Intel Ultra 7 265K CPU, NVIDIA RTX 5070 Ti GPU with 16GB VRAM, and 32GB RAM. The following subsections present the experimental results for spatial-temporal aligned multimodal data generation and machine learning-based antenna pattern reconstruction.

A. Spatial-Temporal Aligned Multimodal Data Generation

In our experiments, we utilize the map of National Yang Ming Chiao Tung University (NYCU) campus for demonstration, which covers an area of approximately 1200 m $\times$ 1200 m, as shown in Figure 2. We simulate the environment from the perspective of a roadside unit (RSU). The spatial-temporal aligned multimodal data generation result is illustrated in Figure 3. The figure shows that we can record both the ray tracing results of the wireless environment and data from various types of sensors in the physical environment such as RGB camera, depth camera and semantic segmentation. In addition to the data types mentioned in the figure, CARLA also supports other sensors such as LiDAR and radar. These high-quality multimodal data provide significant benefits for future research in ISAC and V2X applications which need various data input for cooperative perception.

B. Machine Learning-Based Antenna Pattern Reconstruction

In the antenna pattern reconstruction process, we initialize the antenna pattern using differentiable parameters, enabling the pattern to become trainable. The antenna gain can be

described by $G(\theta, \phi)$, where θ and ϕ represent the zenith angle and the azimuth angle, respectively. The modeling of the antenna pattern should be adapted according to the specific scenario. In our demonstration, we assume that the target antenna pattern to be learned is a dipole pattern, which characterized by constant gain in the azimuth direction. Therefore, we formulate the antenna pattern as follows:

$$G(\theta, \phi) = \left| \sum_{k=0}^{N-1} a_k \cos(k\theta) \right| \quad (1)$$

where N represents the number of cosine basis functions used to approximate the target pattern, and a_k denotes the amplitude coefficient of the k -th cosine basis function.

For this demonstration, we use ray tracing-simulated signal strength data collected from the first floor of a building within the NYCU campus that shown in Figure 4. We configure both TX and RX antennas as dipole antennas and obtain 1,000 valid data points as the dataset. Assuming the TX antenna pattern as unknown and treating it as the target pattern to be learned, we initialize the antenna in the ray tracing environment using the pattern that mentioned in equation (1) with $N = 4$ and execute the reconstruction approach described in section III. The coefficients are initialized with $a_0 = 0.5$ and $a_{k>0} = 0.1$. We adopt the Adam optimizer with a learning rate of 0.01, a batch size of 64, and train for 100 epochs. The model is trained to minimize the mean square error (MSE) between the simulated received power from target and reconstruction antenna pattern.

Table II illustrates the training progress of the antenna pattern. Starting from an initial shape that significantly differs from the target pattern, the trainable pattern gradually converges and closely approximates the target pattern at epoch 100. As the result, the trained pattern achieves 1.6 dB RMSE in the indoor training scenario and maintains comparable accuracy approximately 3 dB when deployed to different outdoor environments such as the NYCU campus map. This result demonstrates significant potential for practical applications where detailed antenna specifications are difficult to obtain. While this work uses a dipole pattern as a proof of concept, the method is not limited to this case. Adapting to more complex patterns, such as directional or MIMO antennas, requires only replacing the antenna pattern formulation with a more suitable mathematical model for training.

C. Analysis of Computation Time

In this section, we discuss the computation time required for system simulation under different conditions. In our system, the total execution time comprises physical environment simulation, ray tracing simulation, communication latency, and parameter updates. Among these components, physical environment simulation and ray tracing simulation constitute the major portions. The following analysis focuses on these two aspects.

First, we examine the physical environment simulation by measuring the time required to simulate scenarios ranging from 1 to 40 vehicles in the NYCU environment. Each simulation step corresponds to 0.1 seconds in the virtual world. Figure 5 presents the average computation time per step in a simulation period of 60 seconds. The results show that if only one vehicle in the map, each timestamp requires merely 9 ms for simulation. When the number of vehicles increases to 40, the simulation time per timestamp increases to 16 ms. It demonstrate that the physical environment simulation achieves near real-time performance even under complex scenarios.

Next, we analyze the computation time for ray tracing in the same NYCU environment. We assume that both the Tx and Rx consist of only one antenna. The ray tracing simulation includes line-of-sight (LOS), reflection, diffraction, and refraction with operating at 28 GHz.

Figure 6a demonstrates the ray tracing computation time for different numbers of Tx and Rx when the ray tracing depth is set to 5. The results show that with only one Tx and one Rx, the average computation time is less than 30 ms. As the numbers of Tx and Rx increase, even with a combination of 8 Tx and 100 Rx, the computation time remains around one second. These findings indicate that GPU-accelerated ray tracing can achieve near real-time simulation in small-scale environments. Depth is also a critical factor that affects simulation accuracy. Figure 6b illustrates the impact of depth on computation time in a single Tx scenario. The figure reveals that depth has a relatively small effect on computation time. Therefore, with sufficient VRAM, we can increase the ray tracing depth to enhance simulation accuracy while maintaining computation efficiency.

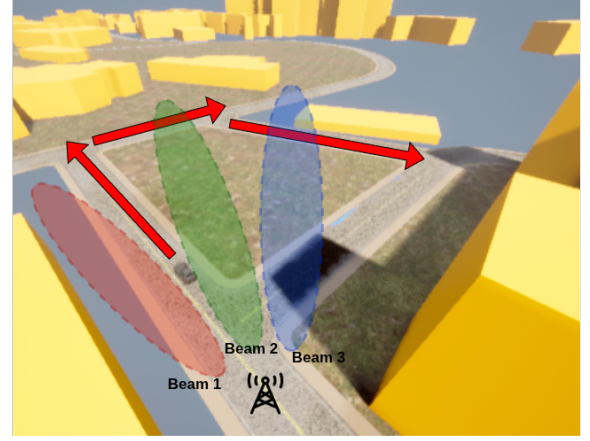


Figure 7: Use case for auto beam selection.

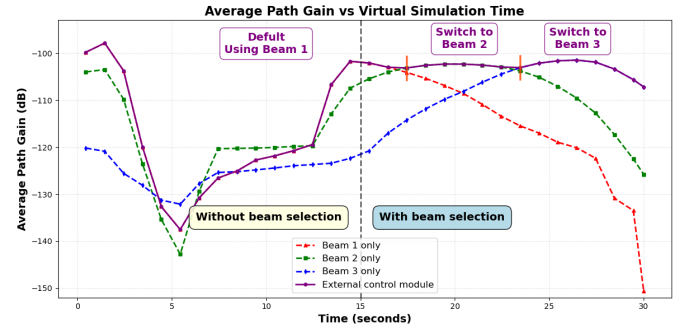


Figure 8: Simulation path gain results of baseline and auto beam selection.

V. Use Case: Closed-Loop Beam Selection via External Control Module

We implement a use case to validate the extensibility and usability of PRISM's context-driven architecture. We consider a beam selection scenario that enhances the path gain of a mobile user through proper beam selection. Experimental setup is illustrated in Figure 7, consisting of a RSU and a vehicle. The RSU has three beam patterns that maintain same shapes but different orientations to select for communication with the vehicle. Each beam is operating at carrier frequency of 28 GHz with a half power beamwidth of 30° in both zenith and azimuth direction. The three beams are separated by 15° in azimuth to provide coverage across different areas. The vehicle moves along the direction indicated by the red arrow with an isotropic RX antenna mounted at its front.

In this use case, we implement an external control module for beam selection. The core simulation workflow remains unchanged, meaning that the physical environment updates the vehicle state and forwards it to the wireless environment for ray tracing simulation as before. The only difference is that wireless environment periodically transmits the position information of both the TX and RX antennas to the external control module via the context-driven interface. The external control module then determines the proper beam pattern

based on the position information and returns the decision to the wireless environment, achieving runtime reconfiguration without modifying the core simulation workflow.

Figure 8 demonstrates the simulation path gain result of baseline beams and external control for beam selection. It is simulated for 30 seconds in the virtual environment. To compare the performance of beam selection and default strategies, the external module maintains the default strategy using beam 1 for the first 15 seconds, while beam selection strategy is activated for the later 15 seconds to choose the proper beam. At the first 15 seconds, it can be observed that beam 1 is not the optimal beam for half of the duration. After activating the beam selection strategy, the external control module sends commands to switch the beam pattern to beam 2 at 17 seconds and beam 3 at 23 seconds. The results show that path gain increases after each beam switching control, maintaining the best path gain performance in the whole duration. These results demonstrate that our digital twin has capabilities for closed-loop simulation and optimization.

VI. Conclusion

This paper presents PRISM, a physics-radio integrated simulation platform for V2X digital twin applications. The platform integrates physics-based vehicle motion simulation with ray tracing wireless propagation through a modular co-simulation architecture, achieving spatial-temporal synchronization and near real-time performance in lightweight scenarios. The closed-loop beam selection use case shows that external optimization modules can be seamlessly integrated to execute context-aware decision making without modifying the simulation workflow, demonstrating the extensibility and usability of MCP-inspired context-driven architecture for future deployment. To address the sim-to-real gap in wireless environment, we introduce a machine learning-based antenna pattern reconstruction method leveraging differentiable ray tracing. Experimental results demonstrate that the reconstructed patterns have good generalization ability across different environments, enabling the platform's effectiveness for realistic V2X and ISAC scenario simulation.

Future works will focus on the following directions. (1) Validation with real-world measurements to verify simulation fidelity of antenna pattern reconstruction. (2) Development of LLM-based agentic modules for autonomous optimization through natural language-based control.

References

- [1] A. Saeizadeh, M. Tehrani-Moayyed, D. Villa, J. G. Beattie Jr, P. Johari, S. Basagni, and T. Melodia, "Airmap-ai-generated radio maps for wireless digital twins," *arXiv preprint arXiv:2511.05522*, 2025.
- [2] A. Zubow, Y. Pilz, S. Rösler, and F. Dressler, "Ns3 meets sionna: Using realistic channels in network simulation," *arXiv preprint arXiv:2412.20524*, 2024.
- [3] J. Borges, F. Bastos, I. Correa, P. Batista, and A. Klautau, "Caviar: Co-simulation of 6g communications, 3d scenarios and ai for digital twins," *IEEE Internet Things J.*, 2024.
- [4] J. Gong, Q. Yu, T. Li, H. Liu, J. Zhang, H. Fan, D. Jin, and Y. Li, "Scalable digital twin system for mobile networks with generative ai," in *Proc. 21st Annu. Int. Conf. Mobile Systems Appl. Services (MobiSys)*, 2023, pp. 610–611.
- [5] R. Pegurri, F. Linsalata, E. Moro, J. Hoydis, and U. Spagnolini, "Toward digital network twins: Integrating sionna rt in ns-3 for 6g multi-rat networks simulations," in *IEEE INFOCOM WKSHPS*. IEEE, 2025, pp. 1–6.
- [6] S.-M. Tang, C.-H. Hsu, Z. Tian, and X. Su, "An aerodynamic, computer vision, and network simulator for networked drone applications," in *Proc. 27th Annu. Int. Conf. Mobile Comput. Netw. (MobiCom)*, 2021, pp. 831–833.
- [7] T. Mao, L. Liang, J. Yang, H. Ye, S. Jin, and G. Y. Li, "Multimodal-wireless: A large-scale dataset for sensing and communication," *arXiv preprint arXiv:2511.03220*, 2025.
- [8] 3GPP, "Study on integrated sensing and communication," Technical Report TR 22.837, 2022. [Online]. Available: <https://www.3gpp.org/dynareport/22837.htm>
- [9] Nokia, "Digital twins in 6g networks," Nokia Bell Labs White Paper, 2025. [Online]. Available: <https://www.nokia.com/asset/215136/>
- [10] H. Luo, S. Jiang, S. R. Khosravirad, and A. Alkhateeb, "Digital twin aided massive mimo csi feedback: Exploring the impact of twinning fidelity," *IEEE Trans. Commun.*, 2025.
- [11] J. Hoydis, S. Cammerer, F. Ait Aoudia, M. Nimier-David, L. Maggi, G. Marcus, A. Vem, and A. Keller, "Sionna," 2022. [Online]. Available: <https://nvlabs.github.io/sionna/>
- [12] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, "CARLA: An open urban driving simulator," in *Proc. 1st Annu. Conf. Robot Learn.*, 2017, pp. 1–16.
- [13] Model Context Protocol Authors. (2024) Introduction to the model context protocol (mcp). LF Projects, LLC. Accessed: 2025-01-21. [Online]. Available: <https://modelcontextprotocol.io/docs/getting-started/intro>
- [14] V. Hassija, V. Chamola, R. De, S. Das, A. Chakrabarti, K. S. Sangwoan, and A. Pandey, "A survey on digital twins: Enabling technologies, use cases, application, open issues and more," *IEEE J. Sel. Areas Sensors*, 2024.
- [15] L. U. Khan, Z. Han, W. Saad, E. Hossain, M. Guizani, and C. S. Hong, "Digital twin of wireless systems: Overview, taxonomy, challenges, and opportunities," *IEEE Commun. Surv. Tutor.*, vol. 24, no. 4, pp. 2230–2254, 2022.
- [16] ns 3, "ns-3 network simulator," <https://www.nsnam.org/>, accessed: 2025-06-26.
- [17] S. Shah, D. Dey, C. Lovett, and A. Kapoor, "Airsim: High-fidelity visual and physical simulation for autonomous vehicles," in *Field Service Robot. (FSR)*. Springer, 2017, pp. 621–635.
- [18] MathWorks, "Ray tracing," <https://www.mathworks.com/help/antenna/ref/rfprop.raytracing.html>, accessed: 2025-06-26.
- [19] Remcom, "Wireless insite 3d wireless prediction software," <https://www.remcom.com/wireless-insite-em-propagation-software/>, accessed: 2025-06-26.
- [20] R. Kumahia, U. Demir, S. Pradhan, B. Salehihi Kouei, K. Chowdhury, and S. Ioannidis, "Ditto: Digital twins for testing and optimizing wireless decisions," in *IEEE Int. Conf. Metaverse Comput. Netw. Appl. (MetaCom)*. IEEE, 2024, pp. 121–128.
- [21] M. Zhu, L. Cazzella, F. Linsalata, M. Magarini, M. Matteucci, and U. Spagnolini, "On the digitization of the em environment: A comparison of ray launching solutions," in *IEEE Int. Mediterr. Conf. Commun. Netw. (MeditCom)*. IEEE, 2024, pp. 85–90.
- [22] Z. Li, R. He, M. Yang, Z. Qi, Z. Zhang, B. Ai, H. Zhang, J. Han, and J. Fan, "Ray-tracing calibration based on improved particle swarm optimization in 5.9 ghz outdoor scenario," *IEEE Antennas Wirel. Propag. Lett.*, 2025.
- [23] C. Yi, W. Chen, Q. Wu, and H. Wang, "Machine learning-assisted calibration for ray-tracing channel simulation at centimeter-wave and millimeter-wave bands," *IEEE Antennas Wirel. Propag. Lett.*, vol. 23, no. 5, pp. 1623–1627, 2024.
- [24] O. Kanhere, H. Poddar, and T. S. Rappaport, "Calibration of nyuray for ray tracing using 28, 73, and 142 ghz channel measurements conducted in indoor, outdoor, and factory scenarios," *IEEE Trans. Antennas Propag.*, 2024.
- [25] G. Gougeon, F. Munoz, Y. Corre, and R. d'Errico, "Ray-tracing calibration from channel sounding measurements in a millimeter-wave industrial scenario," in *18th Eur. Conf. Antennas Propag. (EuCAP)*. IEEE, 2024, pp. 1–5.
- [26] OpenStreetMap contributors, "Planet dump retrieved from <https://planet.osm.org>," <https://www.openstreetmap.org>, 2017.